

EXP 12- IRIS FLOWER CLASSIFICATION

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AIM

To implement the K-Nearest Neighbors (KNN) algorithm for classifying Iris flowers into three species—Setosa, Versicolor, and Virginica—using features such as sepal length, sepal width, petal length, and petal width.

PROCEDURE

1. Load the Dataset: Import the Iris dataset containing flower measurements and their corresponding species labels.
2. Preprocess the Data: Separate the input features (sepal length, sepal width, petal length, petal width) and target classes, and split the dataset into training and testing sets.
3. Train the KNN Model: Select an appropriate value of K (number of nearest neighbors) and train the KNN classifier using the training data.
4. Test and Predict: Use the trained model to predict the species of Iris flowers in the test dataset.
5. Evaluate Performance: Measure the model's accuracy and analyze the results using a confusion matrix and classification report to determine the effectiveness of the classifier.

CODE:

```
import pandas as pd

from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import LabelEncoder

data = {
    'SepalLength':[5.1,4.9,6.2,5.9,5.0,6.7,6.3,5.5,6.0],
    'SepalWidth':[3.5,3.0,2.8,3.0,3.4,3.1,2.9,2.3,2.7],
    'PetalLength':[1.4,1.4,4.8,5.1,1.5,4.7,5.6,4.0,5.1],
```

```
'PetalWidth':[0.2,0.2,1.8,1.8,0.2,1.5,1.8,1.3,1.6],  
'Species':['Setosa','Setosa','Versicolor','Virginica','Setosa',  
           'Versicolor','Virginica','Versicolor','Virginica']  
}
```

```
df = pd.DataFrame(data)
```

```
X = df.iloc[:,0:4]
```

```
le = LabelEncoder()
```

```
y = le.fit_transform(df['Species'])
```

```
model = KNeighborsClassifier(n_neighbors=3)
```

```
model.fit(X,y)
```

```
flower = [[5.2,3.4,1.5,0.2]]
```

```
prediction = model.predict(flower)
```

```
print("Predicted Flower:", le.inverse_transform(prediction)[0])OUTPUT:
```

OUTPUT:

```
Predicted Flower: Setosa  
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but KNeighborsClassifier does  
warnings.warn(
```

(II) import pandas as pd

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.preprocessing import LabelEncoder
```

```
data = {
```

```

'SepalLength':[5.0,5.4,6.0,6.4,6.9,6.3,5.8,6.5,5.7],
'SepalWidth':[3.6,3.9,2.9,3.2,3.1,3.3,2.7,3.0,2.8],
'PetalLength':[1.4,1.7,4.5,5.3,5.4,6.0,4.1,5.5,4.5],
'PetalWidth':[0.2,0.4,1.5,2.3,2.1,2.5,1.0,1.8,1.3],
'Species':['Setosa','Setosa','Versicolor','Virginica','Virginica',
           'Virginica','Versicolor','Virginica','Versicolor']
}

```

```
df = pd.DataFrame(data)
```

```
X = df.iloc[:,0:4]
```

```
le = LabelEncoder()
```

```
y = le.fit_transform(df['Species'])
```

```
model = KNeighborsClassifier(n_neighbors=3)
```

```
model.fit(X,y)
```

```
flower = [[5.9,3.0,4.2,1.5]]
```

```
prediction = model.predict(flower)
```

```
print("Predicted Flower:", le.inverse_transform(prediction)[0])
```

OUTPUT:

```

Predicted Flower: Versicolor
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but KNei
warnings.warn(

```

(III)

```
import pandas as pd
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.preprocessing import LabelEncoder
```

```
data = {  
    'SepalLength':[4.8,5.2,5.6,6.1,6.7,7.0,5.5,6.4,6.8],  
    'SepalWidth':[3.4,3.5,2.9,2.8,3.0,3.2,2.4,2.8,3.0],  
    'PetalLength':[1.6,1.5,3.6,4.7,5.0,5.8,3.8,5.6,5.5],  
    'PetalWidth':[0.2,0.2,1.3,1.2,1.7,1.8,1.1,2.1,2.1],  
    'Species':['Setosa','Setosa','Versicolor','Versicolor','Virginica',  
              'Virginica','Versicolor','Virginica','Virginica']  
}
```

```
df = pd.DataFrame(data)
```

```
X = df.iloc[:,0:4]
```

```
le = LabelEncoder()
```

```
y = le.fit_transform(df['Species'])
```

```
model = KNeighborsClassifier(n_neighbors=3)
```

```
model.fit(X,y)
```

```
flower = [[6.6,3.0,5.2,2.0]]
```

```
prediction = model.predict(flower)
```

```
print("Predicted Flower:", le.inverse_transform(prediction)[0])
```

OUTPUT:

```
Predicted Flower: Virginica
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but KNeighborsClassifier
  warnings.warn(
```

(IV) CODE

```
import pandas as pd
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.preprocessing import LabelEncoder
```

```
data = {
    'SepalLength':[5.1,4.7,5.3,5.8,6.2,6.5,5.9,6.3,7.1],
    'SepalWidth':[3.8,3.2,3.7,2.7,2.9,3.0,3.0,2.5,3.0],
    'PetalLength':[1.5,1.3,1.5,4.1,4.3,5.2,4.2,5.0,5.9],
    'PetalWidth':[0.3,0.2,0.2,1.0,1.3,2.0,1.5,1.9,2.1],
    'Species':['Setosa','Setosa','Setosa','Versicolor','Versicolor',
              'Virginica','Versicolor','Virginica','Virginica']
}
```

```
df = pd.DataFrame(data)
```

```
X = df.iloc[:,0:4]
```

```
le = LabelEncoder()
```

```
y = le.fit_transform(df['Species'])
```

```
model = KNeighborsClassifier(n_neighbors=3)
```

```
model.fit(X,y)
```

```
flower = [[5.7,2.8,4.1,1.3]]
```

```
prediction = model.predict(flower)
```

```
print("Predicted Flower:", le.inverse_transform(prediction)[0])
```

OUTPUT:

```
Predicted Flower: Versicolor
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but KNeighborsClassifier has feature names
  warnings.warn(
```

(V)

CODE:

```
import pandas as pd
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.preprocessing import LabelEncoder
```

```
data = {
    'SepalLength':[4.9,5.0,5.5,5.7,6.0,6.7,6.3,6.8,7.2],
    'SepalWidth':[3.1,3.5,2.5,2.8,2.2,3.0,2.7,3.2,3.6],
    'PetalLength':[1.5,1.3,4.0,4.1,4.0,5.0,4.9,5.9,6.1],
    'PetalWidth':[0.1,0.3,1.3,1.3,1.0,1.7,1.8,2.3,2.5],
    'Species':['Setosa','Setosa','Versicolor','Versicolor','Versicolor',
              'Virginica','Virginica','Virginica','Virginica']
}
```

```
df = pd.DataFrame(data)
```

```
X = df.iloc[:,0:4]
```

```
le = LabelEncoder()
```

```
y = le.fit_transform(df['Species'])
```

```
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X,y)
```

```
flower = [[7.0,3.2,6.0,2.2]]
```

```
prediction = model.predict(flower)
```

```
print("Predicted Flower:", le.inverse_transform(prediction)[0])
```

OUTPUT:

```
Predicted Flower: Virginica
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but KNeighborsClassifier does
  warnings.warn(
```